

Computer Networks Complete Core Notes

A Professional Reference Guide covering Hubs, Switches, DHCP, IP Addressing Classes, and Subnetting Fundamentals

1. Hub

Definition

A Hub is a basic, non-intelligent hardware device used in computer networking to connect multiple devices within a Local Area Network (LAN), acting as a centralized connection point.

Key Features

- **OSI Layer:** Operates strictly at the **Physical Layer (Layer 1)** of the OSI Model.
- **Addressing:** Does not possess any capability to read, process, or store MAC addresses.
- **Data Transmission:** Uses a **Broadcast** mechanism exclusively; incoming data packets are cloned and transmitted to every single connected port.
- **Intelligence:** Lacks routing or filtering intelligence, meaning it cannot distinguish between intended recipients.
- **Performance Impact:** Shares bandwidth among all ports, creating a single large collision domain and broadcast domain, leading to elevated network traffic and data collisions.

Operational Logic

When an endpoint (e.g., PC-A) transmits data through a Hub:

1. The Hub receives the physical electrical signal on the incoming port.
2. Without analyzing the packet frames, the Hub blindly regenerates and forwards the signal to **all other active ports**.
3. Every connected device receives the packet. The network interface card (NIC) of each device inspects the destination address; the intended device processes the frame, while all other devices discard it.

2. Switch

Definition

A Switch is an intelligent networking device that connects various endpoints within a Local Area Network (LAN) and uses packet switching to forward data selectively to its destination.

Key Features

- **OSI Layer:** Operates primarily at the **Data Link Layer (Layer 2)**.
- **Addressing:** Inspects hardware addresses and maintains a dynamic mapping in an internal **MAC Address Table** (or CAM Table).
- **Data Transmission:** Supports efficient **Unicast** communication alongside broadcast and multicast operations.
- **Performance Impact:** Each physical port represents an isolated collision domain, significantly reducing collisions and elevating total available throughput.
- **Security & Efficiency:** Data frames are kept private from unrelated nodes, rendering it faster and vastly more secure than a hub.

Operational Logic & Address Learning

When data frames navigate through a Switch:

- **Initial Ingestion (Unknown Destination):** If a frame arrives for a destination MAC address not yet logged in the MAC table, the Switch floods the frame to all ports except the source port. Simultaneously, it extracts the *source MAC address* and maps it to the incoming port number within its memory.
- **Steady State (Known Destination):** For subsequent frames, the Switch queries its CAM table, identifies the precise port associated with the destination MAC address, and forwards the frame via **Unicast** strictly to that port.

Example: MAC Address Table Mapping

Hardware MAC Address	Physical Port Number
AA:11:XX:XX:XX:01	Port 1
BB:22:XX:XX:XX:02	Port 2
CC:33:XX:XX:XX:03	Port 3

Scenario Application

If node **AA:11** transmits a data packet directed explicitly to node **BB:22**, the Switch parses the destination header, locates **BB:22** matching **Port 2** inside the table, and transparently routes the frame solely to Port 2.

3. OSI Model Layering & Device Map

The table below summarizes the theoretical layers of the Open Systems Interification (OSI) framework and the primary operational hardware mapped to each layer:

Layer ID	Layer Name	Primary Hardware / Security Devices Used
Layer 7	Application	Gateways, Next-Gen Firewalls (NGFW), Proxies
Layer 6	Presentation	Gateways, Cryptographic Offloaders
Layer 5	Session	Gateways, SOCKS Firewalls
Layer 4	Transport	Stateful Firewalls, Layer 4 Load Balancers
Layer 3	Network	Routers, Layer 3 Switches
Layer 2	Data Link	Network Switches, Bridges, Wireless Access Points (WAPs)
Layer 1	Physical	Network Hubs, Repeaters, Modems, Cabling infrastructure

4. DHCP (Dynamic Host Configuration Protocol)

Definition

DHCP is a standardized network protocol used on Internet Protocol (IP) networks for dynamically and automatically distributing essential network configuration parameters to endpoints.

It eliminates manual configuration by automatically leasing the following parameters:

- **Unique IP Address**
- **Subnet Mask** (defines network boundaries)
- **Default Gateway** (Router IP)
- **DNS Server Addresses** (Name resolution)

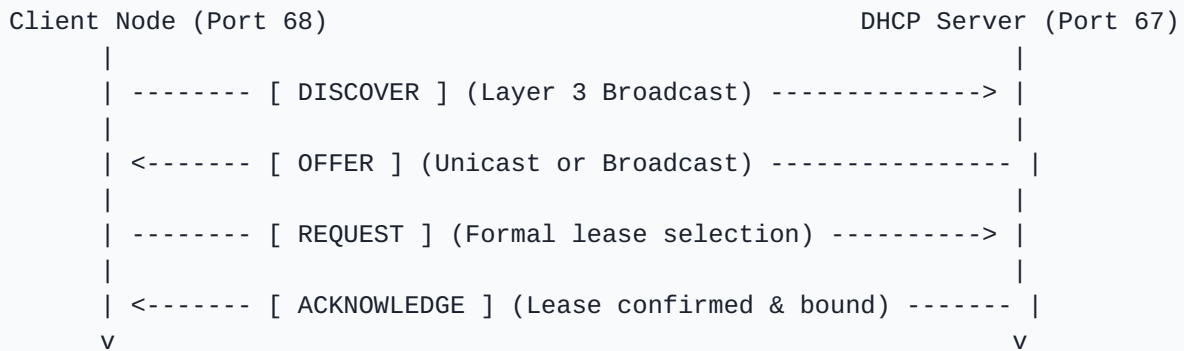
Protocol Port Allocation

DHCP operates over the User Datagram Protocol (UDP) transport framework using specific well-known ports:

- **DHCP Server Port:** UDP Port **67** (listens for client broadcasts)
- **DHCP Client Port:** UDP Port **68** (receives configuration parameters)

The DORA Process Workflow

The sequence diagram below visualizes the standard four-step handshake executed when a new node attaches to an ip network:



1. **D - Discover:** The client broadcasts a DHCPDISCOVER frame across the local network segment requesting an available allocation.
2. **O - Offer:** A listening DHCP server responds with a DHCPOFFER message, reserving and presenting an unassigned IP address alongside standard option configurations.
3. **R - Request:** The client responds with a DHCPREQUEST broadcast, notifying all listening servers that it has formally accepted this specific address offer.
4. **A - Acknowledge:** The chosen server completes the handshake via a DHCPACK message, committing the configuration variables to database storage and binding the lease parameters to the client NIC.

Automatic Private IP Addressing (APIPA)

When a configured client fails to locate a functional DHCP server and cannot negotiate a lease, operating systems like Microsoft Windows engage **APIPA** to self-assign a fallback network configuration. This allows isolated machines on a shared physical switch to communicate without infrastructure routing.

- **Reserved APIPA Network Range:** *169.254.0.1* to *169.254.255.254*
- **Default Applied Subnet Mask:** *255.255.0.0* (or CIDR notations of */16*)

5. IPv4 Class Architecture

An IPv4 address consists of 32 bits arranged into 4 distinct 8-bit blocks called octets. The binary configuration of the **First Octet** structurally defines the default network class boundaries:

Class	First Octet Range	Default Subnet Mask	CIDR Notation	Primary Use Case / Scope
Class A	1 – 126	255.0.0.0	/8	Massive enterprise/infrastructure networks (few networks, vast host counts)
-	127	255.0.0.0	/8	Reserved for Loopback Diagnostics (e.g., 127.0.0.1 for local host verification)
Class B	128 – 191	255.255.0.0	/16	Medium-to-large corporate organizations
Class C	192 – 223	255.255.255.0	/24	Small business, home office networks, localized LAN infrastructure
Class D	224 – 239	Undefined	Undefined	Multicast Group Routing (one-to-many physical streaming data paths)
Class E	240 – 255	Undefined	Undefined	Experimental / Research (Reserved for technical developments)

6. Subnetting

Core Concept

Subnetting is the practice of strategically segmenting a large single logical network space into smaller, distinct, isolated sub-networks (subnets). This is done by borrowing host bits and reallocating them as network identifier bits.

Key Benefits

- **Enhanced Security:** Isolates sensitive corporate departments into distinct broadcast environments, allowing access control lists (ACLs) to regulate inter-subnet data flows.
- **Reduced Congestion:** Minimizes overall network broadcast traffic by confining packets to explicit localized broadcast segments.
- **IP Allocation Efficiency:** Maximizes utilization of finite IPv4 address pools, preventing massive address waste.

Variable Length Subnet Mask (VLSM) / CIDR Table

The matrix below demonstrates structural values resulting from incremental host-bit borrowings within standard Class C spacing blocks:

CIDR	Calculated Subnet Mask	Total Block Size	Usable Host IP Range Formula
/25	255.255.255.128	128	Block Size - 2 = 126 usable hosts
/26	255.255.255.192	64	Block Size - 2 = 62 usable hosts
/27	255.255.255.224	32	Block Size - 2 = 30 usable hosts
/28	255.255.255.240	16	Block Size - 2 = 14 usable hosts
/29	255.255.255.248	8	Block Size - 2 = 6 usable hosts
/30	255.255.255.252	4	Block Size - 2 = 2 usable hosts (Ideal for point-to-point links)
/31	255.255.255.254	2	Specialized point-to-point links (RFC 3021 context)

Mathematical Note on Subnet Values

The fundamental masked values inside the octets (**128, 192, 224, 240, 248, 252, 254**) represent the cumulative sum of high-order binary bits transitioned systematically from an unmasked 0 to a active network mask value of 1.

7. Technical Interview Q&A Reference

Q1: Explain the primary engineering differences separating a Network Hub from a Network Switch.

A Hub functions at OSI Layer 1, possesses no data storage or tracking logic, and broadcasts incoming frames indiscriminately to every port, causing high traffic and collision rates. Conversely, a Switch operates at OSI Layer 2, actively builds and consults a dynamic MAC Address Table, and routes traffic via private Unicast frames directly to the intended device, isolating collision domains and optimizing performance.

Q2: Identify the precise application port mappings utilized across standard DHCP routines.

DHCP relies on the UDP transport protocol. Network DHCP servers listen for and ingest requests using UDP port **67**, while endpoint network clients receive configurations and transmit on UDP port **68**.

Q3: What exact scope limits an APIPA allocation, and when does it trigger?

The APIPA operational spectrum is strictly defined between **169.254.0.1** and **169.254.255.254**. It triggers automatically when a host machine is configured for dynamic leasing but fails to establish communication or capture a lease from a responding DHCP server.

Q4: Define the specific network value and use case assigned to IPv4 address 127.0.0.1.

Address **127.0.0.1** is designated as the local **Loopback Address**. It allows system administrators and applications to test local TCP/IP stack configurations directly on the host machine without engaging physical network adapters or infrastructure.

Q5: Which primary device routes packets across separate logical networks at Layer 3 of the OSI model?

The **Router** is the primary device that operates at Layer 3 (Network Layer) of the OSI model. It analyzes destination IP addresses to determine the optimal paths for routing data across disparate logical networks.

Q6: Which device tracks, logs, and acts upon MAC physical addresses?

The **Switch** tracks hardware addresses. It relies on its internal MAC address table to direct inbound frames efficiently to their precise destinations.